

When Signals Stop Working

A benchmark-relative framework for distinguishing descriptive indicator events from stable incremental predictive value.

Core rule: a signal cannot be said to have stopped working unless it first demonstrated value beyond a common non-indicator benchmark.

V1 evidence architecture

Layer	Public question	V1 method
1. Event	What followed the conventional threshold?	Signed event study after assumed costs.
2. Increment	Does the signal improve market-information forecasts?	Regularized logistic baseline vs candidate, expanding windows.
3. State	Where does the contribution survive?	Training-only Gaussian Markov forward filter.
4. Failure	Has an established contribution deteriorated?	One-sided CUSUM plus recent predictive and economic gates.

Signals and outputs

RSI and Bollinger Bands receive independent verdicts. The combined model is a secondary complementarity test. The output suite includes probability calibration, fold evidence, regime matrices, cumulative benchmark-relative evidence, rolling edge, and structural-change monitoring.

Status taxonomy

NOT ESTABLISHED	Incremental value was not demonstrated; no prior edge is presumed.
ACTIVE	Established value remains positive and no deterioration alarm is active.
REDUCED	Evidence is conditional, uncertain, or deteriorating.
SUSPENDED	Established value crossed structural and recent-performance failure gates.

Methodology roadmap

V1 is the simplest defensible implementation. Later versions add complexity only when it improves locked out-of-sample evidence, uncertainty quantification, or operational control.

Version	Methodological extension
V2	Locked holdout, exchange replication, calibration, Diebold-Mariano and Superior Predictive Ability tests, run-vintage controls.
V3	Time-varying signal coefficients, fully estimated hidden regimes, duration dependence, and model averaging.
V4	Bayesian online changepoints, run-length posterior, and horizon-specific failure hazard.
V5	Funding, open interest, liquidations, order-book state, cross-market transmission, and production governance.

Sources and reproducibility frame

Indicator definitions follow Wilder (1978) and John Bollinger's official material. Regime filtering is motivated by Hamilton (1989). Sequential monitoring follows Page (1954). Forecast comparison and future inference upgrades draw on Diebold-Mariano (1995), Hansen (2005), and Adams-MacKay (2007). Public OHLCV access uses CCXT. Complete bibliographic details and links are maintained in docs/REFERENCES.md.

Research boundary

The framework is venue-, symbol-, timeframe-, sample-, and cost-specific. It is not a trading recommendation, live execution system, or universal claim about technical analysis.